

Multigenerational classrooms in higher education: equity and learning with technology

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Abstract

Purpose – The purpose of this paper is to discuss potential challenges learners from different generations may have with current instructional methods using educational technologies in the classroom. The authors hope to create awareness to help improve equity in learning opportunities and assist educators in understanding the needs of multigenerational classrooms.

Design/methodology/approach – In a narrative review of the literature, the authors present the current findings of the literature on generations in higher education and concerns for equity in learning opportunities.

Findings – It is commonplace in undergraduate programs for learners of multiple generations to attend classes together and research has shown that historical context and generational experiences affect the values, attitudes and learning preferences of each generation. Therefore, higher education institutions should be aware of the demographic profile of their students, as well as the external populations from which they may recruit students, to ensure they are cognizant of the needs of these populations and can provide equality in learning opportunities.

Practical implications – To assist with the needs of this changing student population, university leaders must consider generational characteristics to ensure equity in learning opportunity. Specifically, university leaders and educators in the classrooms will need to adapt and adjust for a changing student population providing instruction that meets the needs of multiple generations of learners, often within one classroom.

Originality/value – Often when we think of diversity in the classroom we think of age, gender, race or even culture. Today we must add diversity in generations. Unlike other equity issues in education such as access (McLaughlin, 2010), educators may not be considering the equity in the design of their instruction to provide equitable learning experiences based on a learners' knowledge and skills established by their experiences with technology. The lack of knowledge and skills a learner has with technology based on their experiences may create barriers to their ability to understand and complete instructional content involving technology (Wager, 2005). To ensure all learners can be successful, educators should strive to provide equality in learning opportunities when designing instruction including technology.

Keywords Equity, Higher education, Learning and technology, Multigenerational learners

Paper type Viewpoint

When examining the current college classroom, it is becoming more common to see the same space inhabited by people from multiple generations (NCES, 2015; Oblinger and Oblinger, 2005; Wager, 2005). Students attending higher education rose 23 percent from 2000 to 2010 and this increase is projected to continue further with another increase of 12 percent by 2025 (NCES, 2015). In a 2017 NCES report of those attending higher education, students 35 and older accounted for 19 percent of the part-time enrollment at public four-year institutions, 34 percent of the part-time enrollment at private nonprofit four-year institutions and 40 percent of the part-time enrollment at private for-profit four-year institutions (NCES, 2017). These participation rates and the trends they show translate directly into classrooms featuring significantly larger populations of older adult students learning alongside their younger peers (Melick, 2014; Wager, 2005).

The diversity of ages in the modern classroom is changing as some people are choosing to enter higher education later in life after having a family or returning to college after years



in the workforce (Melick, 2014; NCES, 2015). Older adults are pursuing an undergraduate degree for a variety of reasons, ranging from personal interests, work-related needs, to economic necessity (O'Donnell, 2005; Parks *et al.*, 2013). Members of each generation exhibit fundamental differences in attitudes, values, and learning preferences as a result of their different historical experiences. The dominant cohorts in today's college classrooms are Baby Boomers (early and late Boomers), Generation X (Gen X), Generation Y (Gen Y or Millennials) and Generation Z (Gen Z, iGen, Post-Millennials, Net Generation, Gen 2020 or Neomillennials) (see Table I).

Often when we think of diversity in the classroom we think of gender, race and culture. Today we must add diversity in generations. The increase in the average age of students represents an expansion in diversity of knowledge, experience and preferences among all students. When there is significant diversity in the ages of learners, there can be challenges in meeting the personal and educational needs of such a heterogeneous student population. Some challenges include meeting the different personal needs for work and family while balancing school. One of the challenges for colleges and educators of multigenerational classrooms is devising the best methods to provide effective and equitable educational experiences to meet learner needs.

Learners of a specific generation are often tied together by their shared experiences with which they grew up, influencing behaviors and expectations (DelCampo *et al.*, 2011; Howe and Strauss, 2007). Wager (2005) stated "Beyond the obvious difference of age and time away from the classroom, adult learners may not have the same comfort level or familiarity with technology – and they may be the least advised on how to use it" (p. 10.6). For example, learners may have had different experiences with the types of technology used in the classroom which may influence their level of comfort with technology later in life (Howe and Strauss, 2007; Lester *et al.*, 2012; Mannheim, 1952). Further, a lack of exposure to technologies, specifically technologies that may be used in a classroom setting, can be especially challenging for learners that did not have exposure to these technologies while in their formative years of schooling and have not been in school for many years. Because of these differences in life experiences, some learners may be at a disadvantage as they may not have the same knowledge and skills as other learners and this may be particularly prevalent when comparing older learners and younger learners. Using instructional methods and technologies unfamiliar to learners of different generations and expecting technology knowledge, skills and comfort may cause some learners to experience significant stress and challenges in their learning.

Problem

Today, the ubiquitous use of technology in the college classroom has led to use of technology for a variety of purposes including communication, coursework, information storage and research. For learners that grew up surrounded by technologies and experiencing it from early years in school, the integration of technology in the learning process may be expected (Greig, 2009). However, for learners that grew up when there was little technology in the classroom, the expectations in current classrooms may be overwhelming. Learners who lack experiences with technology in the classroom, may be at a disadvantage when considering the opportunity to learn with technology as they will have significant differences in their knowledge and skills. Providing access to the same

Baby Boomers		Generation X (Gen X)	Millennials (Gen Y)	Generation Z (Gen Z)
Early Boomers Born 1945–1955	Late Boomers Born 1956–1964	Born 1965–1979	Born 1980–1999	Born 1995–2010

Table I.
Dominant cohorts
in today's
college classrooms

instructional content and technology resources may not mean equitable opportunity (McLaughlin, 2010). Taking generational characteristics into account is important when designing instruction and teaching, as failure to consider the needs of learners from different generations may result in an inability for all learners to be equally successful (DelCampo *et al.*, 2011). Other problems that can result from not considering students' generational characteristics could include students having a less than favorable learning experience and attrition issues. Therefore, we argue that by providing learners with the opportunity to access instructional content using technology does not mean there is equality in learning opportunity. Rather, learners who lack the knowledge and experience of their peers can encounter significant differences in their equity in learning.

Purpose

The purpose of this paper is to discuss potential challenges learners from different generations may have with current instructional methods using educational technologies in the classroom. Although, the authors do recognize that experiences and access to educational technologies may also be a result of socioeconomic status and other known cases of digital access issues, this paper is focused on challenges that are the result of generational experiences. Through writing this paper, the authors hope to create awareness to help improve equity in learning opportunity and assist educators in understanding the needs of the current multigenerational classroom.

Structure

The structure of this paper is as follows: we first present our methodology, including our research questions, then we respond to the questions using the literature, and lastly, we discuss our findings. For the purposes of this paper, we will discuss equity in the higher education environment as striving toward providing students with the opportunity to achieve equal levels of success (McCowan, 2016). While this paper will focus primarily on equity in the USA' higher education system, considerations toward equity in higher education can also be found in other countries (Le and Miller, 2005).

Methodology

In order to address the purpose of this paper, which is to understand the potential challenges learners from different generations may have with current instructional methods using educational technologies in the classroom, we asked the following questions:

- RQ1.* What is equity in education?
- RQ2.* How does digital technology come into play when discussing equity in education?
- RQ3.* Which are the most prominent generations in our college classrooms and what are their learning experiences with technology?
- RQ4.* What are students of different generations finding when they return to school?
- RQ5.* Do people from different generations have equity in their opportunity to learn while using technology in higher education classrooms?

To answer these questions, it is our intention to describe and synthesize the available literature on this topic and provide conclusions from this evidence. In this narrative overview of the literature, we synthesize and discuss the findings of literature we retrieved from searches of online computerized databases, manual searches by hand and authoritative texts. The researchers followed several procedures to ensure a high quality review. First, a search was completed of peer-reviewed journals, conference papers, and reports using a wide range of key terms. The following are some of the key terms used: adult learners, higher education,

equity education, and equity in learning, generation, generation technology use, generational digital divide, digital age gap, generational learning preferences, technology, information age and digital technology. The researchers searched in several databases including: Education Resources Information Center, Education Full Text, ABI/Inform, Business Source Complete, Google Scholar and ProQuest Dissertations and Theses Global. Second, the reference section of each article was reviewed to locate additional useful sources. Third, key education and technology journals and other publications were searched independently including the following publications: *Computers and Technology*, *British Journal of Educational Technology*, *Educational Technology Research and Development*, *New Horizons in Adult Education & Human Resource Development*, and *Training*.

The search process revealed peer-reviewed articles, dissertations, books and reports. Some initial themes that were considered relevant to the topic included generational diversity in learning, adult learners and technology, higher education and adult learners. Once we delved further into the research, we found that generational experiences during formative years of learning and the generational digital divide or technology age gap, were important themes in this research. We followed a two-stage screening process with two independent researchers to determine if the sources found in the search included themes relevant to the topic. Themes that emerged were based on topics discussed within the literature. The following review of the literature will discuss our findings.

What is equity in education?

To begin the conversation about equity across multiple generations in higher education, we first define the idea of equitable learning in education.

Equity in education

Equity in education is often viewed as equality or equivalence (sameness) and is sometimes described as a situation in which everyone is treated fairly and equally (Masters and Adams, 2018). Masters and Adams (2018) pointed out that the implication of this definition of equity is that “equity depends on equality of treatment” (para. 2), when in reality, equality really comes down to fairness of treatment (Anderson and Hendricks, 2014; Masters and Adams, 2018). Fairness relates to what is fair based on what can be provided and what is needed. In education, equity does not mean equal distribution or even equal treatment, as treating everyone the same, or equal, is actually not fair. We have found in our schools and classrooms that there are students that have unequal learning needs (e.g. physical and cognitive disabilities) and unequal backgrounds (socio-economic), which can mean that there are differences in their needs when compared to the majority of students. These differences can mean that for some students additional support is needed. Therefore, “equity deals with accommodating and meeting the specific needs of specific individuals” (Anderson and Hendricks, 2014, p. 8). It is through equity that we are providing each student what they need to be successful.

Equity in education will not always ensure that students are treated the same or that they achieve at the same level, and this illustrates the distinction between what is fair and what is equitable (Unterhalter, 2009). Therefore, learners may be treated differently in order to provide some learners with an equitable learning experience (Unterhalter, 2009).

Thus, we argue that equity in learning opportunity goes beyond simply providing access or equal access. For example, instructors may provide instruction and resources to learners, but a learner who lacks knowledge and skills to use the technology based on their experiences may experience tremendous barriers to their ability to understand and complete instructional content involving technology, thus access without equity. Therefore, we can apply the argument that has evolved within K-12 education to higher education, it is important to acknowledge that only providing access for learners to the content may not offer equitable opportunity and potential for success (McLaughlin, 2010).

How does digital technology come into play when discussing equity in education?

Digital technology

Many students today grow up using different types of technologies at home and in their schooling. Digital technology (also known as digital electronics or digital devices) came about in the mid-twentieth century with the conversion of information into a digital format. The proliferation of digital devices during the Digital Revolution (1950s to late 1970s), marked the beginning of the Information Age (Castells, 1996). These digital devices, mainly computers with the inclusion of software, could create, display, record, analyze and share information. Castells (1996) described the Information Age as a period in which among other things, information can be transferred through networks (Beetham and Sharpe, 2013).

Digital technologies have permeated the daily lives of people and are used in our work, education, time with our families and friends and our alone time. With the invention of the internet and then the World Wide Web (WWW) in 1992, web-based digital devices and applications have been able to facilitate the transfer of information over networks (Castells, 1996). Through using digital technologies, people can create and analyze content and with the WWW, people can share such content. For our use in this manuscript, when we refer to technology, we are referring to digital technology, which includes any hardware, software or other device or tool that can be described as electronic, digital or computerized.

Digital technology in classrooms

Today, a variety of digital technologies are used in classrooms for various purposes including communication, coursework, information storage and research. In some cases, the institution selects the technologies to be used (e.g. e-mail system, learning management system (LMS)), in other cases, it is the instructor or students (e.g. presentation software, web-based applications).

In a study published in 2003, Kozma (2003) examined 174 case studies from 28 countries of innovative pedagogical practices. In this study, Kozma found that students and instructors use technology tools for purposes such as searching for information, designing products, publishing findings and teachers use technology for instructional purposes including presenting instructional content, monitoring student progress. In more recent studies, the types of and purposes for technologies used in classrooms have expanded significantly (Buchanan *et al.*, 2013; Fichten *et al.*, 2015; Venkatesh *et al.*, 2016). Some of the different types of technology used in classrooms for instructional purposes include the internet, e-mail, course/LMSs (e.g. Blackboard, Moodle), word processing, blogs and wikis, social media (e.g. Facebook, Twitter), presentation software (e.g. PowerPoint, Prezi), online audio and video, cloud and web based applications (e.g. Google Drive), virtual and augmented reality software and devices, tablet and SmartPhone apps, interactive devices (e.g. clickers) and interactive whiteboards (e.g. SmartBoard) (Buchanan *et al.*, 2013; Fichten *et al.*, 2015; Venkatesh *et al.*, 2016).

Which are the most prominent generations in our college classrooms and what are their learning experiences with technology?

To begin our discussion of the students from various generations in our classrooms, we will first define what generation means. Then we will review the four most prominent generations represented in our classrooms and their learning experiences with technology.

Defining generation

We first see the term “generation” described in the literature by Karl Mannheim in his writing “The Problem of Generations” which he published in 1952. Mannheim saw that a

generation was tied together by a similar worldview resulting from shared events and mindsets, rather than solely being defined by shared age (Pilcher, 2009). We can see this idea continue with Meredith and Schewe (1994) defining the span of a generational cohort as 20–25 years. These 20–25 years of shared experiences form the foundation of a generation, influencing how these individuals may interact with the world around them and their own education (DelCampo *et al.*, 2011; Howe and Strauss, 2007; Reeves, 2008). Though specific dates may vary according to sources and names of each generation may vary, generational characteristics and values remain consistent (Dwyer and Azevedo, 2016; Greig, 2009; Macky *et al.*, 2008). Although generations are often defined by their age, Reeves (2008) wrote “generations are shaped much more by history than by chronological dates. It is important not to make assumptions about any one person based on birth year” (p. 3).

Moving beyond a strict view of age distinction, there is research indicating that generational distinction influences how we see and interact in the world around us (Cogin, 2012; Lewis and Wescott, 2017; Mannheim, 1952). When we look at how the shared experiences of a generational cohort can influence values and preferences, generation serves as another contributing factor to what shapes who we are (Cogin, 2012; Howe and Strauss, 2007; Lewis and Wescott, 2017; Meredith and Schewe, 1994; Quinn, 2010). Generational differences can be more and less pronounced in certain individuals, leading to those who may not fully identify with their generation. It is also worth noting that differing views remain regarding the extent to the impact of an individual’s generation (Bell, 2008; Costanza *et al.*, 2012). The ability to define a generational cohort allows us to understand the needs, commonalities and differences among members of different generations to consider equity in how we teach an increasingly age-diverse population of learners.

Most prominent generations and their learning experiences with technology

In this section we present the four major generations that are found in college classrooms and a general review of their experience with technology.

The four most prominent generations in our college classrooms today are Baby Boomers (early and late Boomers), Generation X (Gen X), Generation Y (Gen Y or Millennials) and Generation Z (Gen Z, iGen, Post-Millennials, Net Generation, Gen 2020 or Neomillennials) (see Table I). Members of each of these generations has distinct differences in their experiences in life, and specifically, in their experiences with technology used in their education.

Baby Boomers

The Baby Boomer generation in the USA arose after the Second World War, with these individuals having been born between 1946 and 1964 (DelCampo *et al.*, 2011). The Baby Boomer generation is divided into early Boomers (b. 1945–1955) and late Boomers (b. 1956–1964) by several researchers as there are distinct differences in the experiences of members of the generation during these years (e.g. Early Boomers – rise of the counterculture, rallies for social justice for women’s rights, civil and gay rights, and Late Boomers – post-war economy and economic hardship, career concerns) that may impact the characteristics of these groups (D’Amato and Herzfeldt, 2008; Poulos and Nightingale, 1997). While all Baby Boomers saw the launch of the personal computer in the early 1980s, they did not grow up with the same level of exposure to the technology as their children or grandchildren both inside and outside their classroom (Boysen *et al.*, 2016; Cekada, 2012; Geisen, 2009). With significantly less technology exposure, this generation lacks experiences using technology in the classroom and their early career. This can create a challenge as Baby Boomers, like other learners, often draw upon their experiences as a source of knowledge when learning or working (Cekada, 2012). We can see this in that there is a significant difference between Baby Boomers and their use of technology when compared to

other generations. Baby Boomers are less likely to adopt newer or innovative technologies as they tend to be more focused on using technology to accomplish specific tasks in their lives, rather than use it because it is available (Geisen, 2009).

Generation X

Born between 1965 and 1979, Gen Xers were heavily influenced by the rapidly changing personal technology of that period (e.g. home video games; information storage devices such as audio and video cassette recorders, launch of the personal computer) as this technology became more common (Klie, 2012). Growing up with more technologies in their classrooms than Baby Boomers and experiencing opportunities to learn autonomously may explain Gen Xers' preference to direct their learning and their appreciation for flexibility in how they pursue learning (Cekada, 2012). These older Gen Xers also experienced the launch of the WWW in 1993 when they were in their mid-20s, providing the ability to take advantage of the new opportunities presented by the internet's expansion (Greig, 2009; Michael, 2012).

Generation Y

Members of Gen Y (1980–1999) are commonly known as “Millennials” (Delcampo *et al.*, 2011; Howe and Strauss, 2007), but this paper will use the term “Generation Y” or Gen Y (Stevens, 2010). Gen Y was born as the personal computer was first sold in mass quantities to the general public and these individuals are likely to have developed comfort with this technology at an early age (Notarianni *et al.*, 2009). With more than 70 percent of K-13 students using computers in class in the late 1990s (Greig, 2009), this generation is the first generation to experience the computer in their classrooms during their early formative years. The oldest members of Gen Y would also have been in their early teens when the WWW first became available in 1993 (Erickson, 2012). With ubiquitous amounts of easily accessible information on the WWW, members of Gen Y and the more recent, Gen Z, are known to expect instant gratification and quick responses (Cekada, 2012). When we look at how prevalent the internet has become since the older Gen Ys were born, this generation is seen as preferring or even expecting technology to be integrated into their school, work and personal lives (Greig, 2009). Erickson (2012) writes that Gen Y was the first generation to be “unconsciously competent users of both the personal computer and the Internet” (para. 3). Gen Yers may expect technology to be included in instruction and they prefer the freedom to explore topics using various digital courses (Cekada, 2012; Su, 2007).

Generation Z

Gen Z (b. approx. 1993–1995–2009), also known as Net Generation, Neomillennials and Digital Natives, is the first cohort to have internet and digital technologies available from a very young age (Prensky, 2001). A report from Pew Research, found that “92% of teens report going online daily – including 24% who say they go online ‘almost constantly,’ this is likely due to nearly three-quarters of teens having, or having access to, a smartphone” (Lenhart, 2015, p. 1; Lenhart *et al.*, 2015, p. 2). This means this generation is the first to grow up with easy, early and consistent access to the internet. Gen Z is described as having a “digital bond to the Internet” by Turner (2015). According to Sparks and Honey (2014), 41 percent of Gen Zers spend more than 3 h per day using computers for purposes other than schoolwork. In addition to access to more digital technology than any previous generation from an early age, this generation has also benefited from hundreds of thousands of computer and digital games and apps created specifically to educate. These educational games and apps are used at home and in the classroom. In addition to games and apps, other types of technologies not seen previously in classrooms (e.g. social media) have also found a place in the classroom with Gen Zers' preference for social interaction.

What are students of different generations finding when they return to school?

Unlike a few decades ago when the average student attending higher education was a young adult, usually 18–24-years-old, today, people from multiple generations fill college classrooms (NCES, 2015; Oblinger and Oblinger, 2005; Wager, 2005). Adult learners, previously referred to as “non-traditional learners,” are choosing to attend college in large numbers. In fact, the largest increase among students has been adult learners 35 and older (NCES, 2017). As adults of different generations are attending college for the first time, returning to college or earning their first degree (see Table II), and many are finding the classroom is different than they experienced when attending school in their formative years. Today, digital technologies are used in classroom instruction and students are expected to use various digital technologies to complete instructional tasks.

Do people from different generations have equity in their opportunity to learn while using technology in higher education classrooms?

As some learners are returning to school, they are finding digital technologies are used in classroom instruction and they are expected to use various digital technologies to complete instructional tasks. In this section, we will discuss digital divide and the lesser known generational digital divide, also known as the technology age gap and its impact on equity of learning for learners returning to college.

Defining digital divide

The term digital divide is often used to express a disparity between the haves and the have nots, in the areas such as education, industry and politics. The digital divide refers to a gap between those who have access to information and technologies such as computers and the internet and those who do not. Livingstone and Helsper (2007) wrote that the digital divide is created by several factors including age, gender and socioeconomic status. Access is a known determinate of the digital divide (Ching *et al.*, 2005; Wei *et al.*, 2011). Ultimately, access refers to access to technology in homes, schools, businesses and geographic areas. Wei *et al.* (2011), discuss three levels of digital divide with the first being access, the second being usage capability, which can be the result of access divide and the third being the outcome divide, where some students will have less learning due to the divide. Students who have not had access to technology due to socio-economic or other factors, such as the lack of technologies in the classroom due to their inexistence or public distribution, may have not developed the same skills and same outcomes. The third level is outcome, which refers to the

	2011–2012 (%)	2012–2013 (%)	2013–2014 (%)	2014–2015 (%)
Bachelor’s degrees as percentage of all bachelor’s degree earners	78.2	77.3	77.1	76.5
<i>Age group</i>				
Under 25	88.7	88.3	88.1	87.9
25–29	57.6	56.3	55.4	54.2
30–39	50.5	48.6	47.0	44.8
40–49	59.8	57.6	54.6	51.9
50 and over	65.2	62.5	61.6	59.9

Notes: This table shows the percentage of bachelor’s degree recipients in each year and age category who were first-time graduates. First-time graduates are defined as students who earned a bachelor’s degree, associate degree, or certificate as their first postsecondary credential. The statistics come from National Student Clearinghouse Research Center (2016)

Table II.
First-time graduates
earning a degree

differences in outcome and productivity that are a result of having access to and opportunities to use technology. For example, Ching *et al.* (2005) found that students that had access to technology at home before age 10, were much more likely to feel comfortable and use the technology to its full capabilities.

Generation and the digital divide

Although there has been significant research that shows that there are generational differences and preferences based on experience, whether generational membership is a cause for the digital gap for some individuals has only been minimally researched and very few research studies have been completed in higher education. Kim and Kim (2001) investigated potential causes of the digital divide and found that generation is a factor that leads to the digital divide. However, they found that it is one of several factors that can lead to the divide, including gender, generation, class and region. Therefore, an individual that is a member of several of these categories, may experience a larger gap than others.

Although there has been some research (Garcia and Qin, 2007) on what impact the generation factor has on the digital divide and student comfort and use of technologies, it has not been conclusive. For example, when considering whether the generational differences impact potential student comfort or use of technologies, Garcia and Qin (2007) found in their study that it may impact it, however, they found that there were much larger differences between the attitudes toward teaching methodologies for learning among members of different generations. Therefore, student preferences and attitudes toward instructor teaching methodologies, including the choice to use technology, could also be a factor contributing to the digital divide. However, in Garcia's and Qin's (2007) study, students were lumped into four groups, with all older adults (36 years and older) in one of the groups. Grouping the students in this manner would make it hard to know if there were significant differences in Gen Z, Millennial, Gen X or Baby Boomer, use or comfort with technologies in the college classroom.

Bullen *et al.* (2011), found in their comprehensive review of research and popular literature on generational use of technology, and also in their empirical study that "there are no meaningful generational differences in how learners say they use ICTs [Information and Communication Technologies] or their perceived behavioural characteristics" (p. 1). However, we have concerns about this study as Bullen *et al.* (2011) examined generations born after 1982, when the personal computer had become mainstream and the Web was a part of the early lives and formative years of education for each generation born after this date, unlike earlier generations, Generation X and Baby Boomers. In fact, all generations in their study were lumped into either Net Generation or Non-Net Generation, and most of the participants were claimed to be Net Generation (78 percent). Again, grouping students in this manner, and having such a large percentage of Net Gen participants would make it hard to know if the Net Gen, the Millennial or other generations were truly different in their learning preferences and use of technology. In addition, the authors included an empirical study including students at only one post-secondary institution, in 2008 when many Net Gen students were still in high school.

As we begin to examine the technology learning experiences and preferences of each generation, we must consider how the classroom technology available when a generation was first in the K12 environment may impact that generation's preferences for using technology to learn (Howe and Strauss, 2007; Lester *et al.*, 2012; Mannheim, 1952; Turner, 2015). The differences in classroom technology can be seen when comparing how Baby Boomers were taught with little technology to the technology available for Gen X by the time these individuals completed high school (Cuban, 1993; Greig, 2009). The evolution of classroom technology becomes even more apparent when we consider that the educational technology available to Gen Y and Gen Z would have included tablets, smart boards, and personal computers (Greig, 2009). This difference would also have extended to

the technology prevalent outside of the classroom at the time, with Baby Boomers having grown up in a world without the smart technology of today (Electronic Design, 2002; Greig, 2009). Looking solely at what each generation would have developed familiarity with in the K12 environment, this serves to illustrate the differences in what each generation may expect in a typical classroom environment.

The Generational Digital Divide and Learning in Higher Education

It can be challenging for people to understand that age can be a factor of the digital divide. However, as learners bring their knowledge and experiences into the classroom (Knowles, 1980), and this knowledge and experience is important for learner satisfaction and success, older learners may be at a disadvantage if they have not kept up with new technologies as these technologies were invented (McMullen, 2016) creating an age-related divide. Therefore, as access to technology has increased, the digital divide is changing to an inequality based on knowledge and skills (Blau, 2002). The result of this lack of access or inequity is what is known as the generational digital divide. The generational digital divide (also known as the technology age gap) refers to the gap in exposure to technologies by age (Livingstone and Helsper, 2007; Salkowitz, 2008). For institutions of higher education and adult learners, this is troublesome as some adult learners may struggle with technology used in classrooms today, therefore feeling separated from their peers and they may choose to withdraw from their programs. McMullen (2016) writes that unfortunately, “technology has an obsoleting impact on those without the proper skills and, with the speed at which the technology changes, it is very difficult – near impossible for some – to keep current” (p. 1). McCoy (2010) wrote that online adult learners have lower self-efficacy scores when asked about the tools they are using in their courses. The effect of lower self-efficacy is that adult learners are more hesitant to enroll in online classes, they may have lower retention rates and they report lower levels of course satisfaction if they lack technology support (Allen and Seaman, 2013; Beaghan, 2013; Melick, 2014).

Badke (2008) found that adult learners have more difficulty performing basic computer functions as compared to their young classmates. Badke (2008) argues that a barrier for adult learners using technology is making the transition between how things were done previously and how things are done now with digital technology. This aligns with Knowles’ (1980) theory of andragogy, which emphasizes the importance of an adult learner’s previous experiences influencing how that adult will learn; in this case, generational inequity could arise when we assume that all learners share comparable previous experiences to prepare them to enter the classroom at the same levels. Although the numbers of adult learners attending higher education is rising, a large number of these learners may not be proficient with the technology used in classrooms today (McCoy, 2010; Rodriguez *et al.*, 2008). A study completed by McCoy (2010) found that students age 18–25 scored higher than older learners on topics relating to technology proficiency, thus leading to lowering perceived skill level among the older learners. As a result, McCoy (2010) has called for higher education institutions to understand their students’ technology proficiency to ensure student satisfaction and retention.

Discussion

With non-traditional students being drawn from Generation X and Baby Boomers, the possibility of a multi-generational classroom in the traditional higher education setting has increased over the years (NCES, 2015). There are several differences that set generations apart from one another and these are based on the shared experiences and events of particular generations (DelCampo *et al.*, 2011; Howe and Strauss, 2007). These differences carry over naturally into the higher education environment, especially as students are expected to self-regulate their learning. One of the key differences we see between members of generations is the knowledge and experiences with technology and how they interact with technology (DelCampo *et al.*, 2011).

Although equity in education has been primarily focused on racial, socioeconomic, or ability differences (Le and Miller, 2005; McCowan, 2016; McLaughlin, 2010), equality in education should be extended to any key difference that may set groups of learners apart from those who would be considered typical learners. As members from each generation has different experiences with technology, we can surmise that learners without proper knowledge and skills may find it very hard to keep up with the current pace of new technologies used in the classroom (McMullen, 2016). Therefore, although people from different generations may have equal access, there may not be equality in opportunity to learn based on learners' skills to use technology in the classroom. McLaughlin (2010) writes that access to the same content may not mean an equitable learning opportunity. Instructors must consider the generational needs of higher education students as access alone may not be enough to avoid unintentional barriers for students from different generations.

While there is research to support the idea that learners from all generations are accepting of and willing to engage with technology in the classroom (Boysen *et al.*, 2016), instructors cannot assume that all learners will share the same levels of familiarity and comfort with the technology to be used in the classroom. For example, Gen Y has been shown to have a preference to using social media and mobile devices in their education (Boysen *et al.*, 2016; Cekada, 2012; Turner, 2015); assuming that all learners shared this same generational preference and aptitude would naturally place Baby Boomers, Gen Xers, and Gen Zers at a disadvantage for any activity requiring proficiency in these technologies.

Building a more equitable learning environment for the multigenerational classroom begins by acknowledging that a multigenerational classroom is comprised of several different types of learners with different needs. While there is always going to be overlap between different generations, it is vital to remember that instructional strategies may need to be adjusted in order to meet the generational needs of students. This can involve an instructor adjusting the strategies to be flexible with adult learners who may need to learn new technologies and also make best use of the strengths that each generation can bring to the classroom. The recommendations available for designing instruction or teaching technology to a multigenerational audience differ in how they choose to approach a solution. Some authors recommend focusing specifically on a single generation's needs (Cekada, 2012; Turner, 2015), while others advocate for a more general approach for the class as a whole (Cekada, 2012; Geisen, 2009; Turner, 2015). We feel that improving equity in the multigenerational classroom starts with considering the needs of the learners separate from the assumptions that we may hold about prospective students, so that all students have equal opportunity to succeed in the classroom.

What can educators do?

To facilitate generational equity in their course, instructors can begin with assessing their students' characteristics and needs and their course to meet the needs of a multigenerational classroom. The authors have five questions that instructors can use to assist in identifying potential concerns and improving generational equity within their classrooms:

- (1) Do I have a firm understanding of my students including their prior experiences? With a rising trend in the number of older students taking undergraduate college courses, understanding generational characteristics of students in the current classroom may assist educators when working with learners from different generations to provide a more equitable experience for learners from all generations. Considering who our learners are and what they need is nothing new to education (Dick *et al.*, 2015; Hascher, 2013); expanding this to include generational factors allows us to meet the changing student population and learner needs and strengths that might otherwise go overlooked. Perhaps surveying students prior to instruction or asking questions during

instructional time will provide more information about your learners including their experiences and skills. Using the results of such a survey or questions, instructors can implement instructional strategies that meet the needs of their students.

- (2) Have I avoided assumptions of my students' technical skills? An important step in enhancing equity in a multigenerational classroom begins with working to avoid the assumption that all learners have the same technology experiences, familiarity, and skills. Learners who do not have experience with technology in an education context may not have developed a comfort with using technology for education-related tasks. Attending college can be challenging, adding the pressure of learning to use new technology to complete course work when there is a lack of knowledge or experience with technology can create tremendous stress. By avoiding the belief that all learners will be similar in their technology skills, instructors are in a better position to support the diverse multigenerational classroom. To assist with this, instructors should ensure the course prerequisites and syllabi provide learners with guidance on expected technological skills and knowledge and list technologies that will be used in the course. In addition, instructors should provide some resources and support to learners who need them. Instructors can also assist these students to become more comfortable with technology through providing text-based and video tutorials, guides, and information of whom students can contact for technology questions or issues. In addition, as less technology-experienced and skilled learners will likely need more time to learn a new technology, educators can provide flexibility for learning and completion of assignments.
- (3) Do I provide resources for older students who may have age-related challenges with using technology? Although some learners will have physical or learning disabilities, adults may have an increased challenges directly related to aging. Sight-related issues, cognitive decline, challenges with mobility and dexterity are all complications adults may struggle with despite a strong desire to learn and attend college. While these age-related problems can be compounded with other issues, such as the discomfort addressed in Question No. 2, this may not always be the case. Providing alternative activities, patience, flexibility and resources such as connecting with a school disability resource center can assist older adult learners that may have challenges with learning or using technology.
- (4) Do I allow for learner choice to use technology? As learners may prefer another option, allowing learners to choose to use or not use technology can provide students with the opportunity to select into an approach that best fits their preferences (DelCampo *et al.*, 2011). For example, allow students to present their project as a poster rather than a Prezi.
- (5) When possible, do I allow for learner choice for technology selection? It may not always be possible for learners to avoid utilizing technology to some extent, but allowing some choice in the type of technology can provide learners from multiple backgrounds with different ways to play to generational strengths. For example, allow students to choose to create their project using a PowerPoint rather than another software program with a much more challenging learning curve.

Conclusion and future research

This manuscript focuses on the available literature and research that explores both generational characteristics and equity in the classroom. While these topics can intersect meaningfully, there is limited research specifically dedicated to this topic. Additional research is recommended, in order to more fully explore how to best serve different generations that wish to pursue higher education.

As the higher education landscape continues to expand, we can expect that the generational diversity within a given classroom will also expand. New generations will continue to enter higher education, further elaborating upon what we think of as the typical university student. The significance of equity in education lies in providing a diverse set of learners with equitable learning experiences regardless of individual learner background or characteristics, so that all learners have the same chance of being successful (McLaughlin, 2010). This moves beyond simply providing access in the hope that this will bring about equality; awareness in and of itself does not remove barriers to equity, but it can then allow us to work toward eliminating those barriers (McCowan, 2016). This paper seeks to help illustrate the need to think about equity in this multigenerational classroom, to ensure that all learners can succeed.

This paper has analyzed the existing literature in order to demonstrate that generational characteristics can also factor when considering equity in the higher education environment, but additional research is needed. As the generational makeup of the university classroom continues to expand, this will provide many opportunities to examine how multiple generations can best learn in a shared environment. Through conducting the appropriate qualitative and quantitative research on generational characteristics in the higher education environment, future researchers will be positioned to further clarify which strategies best appeal to the diverse generational needs of a changing student body.

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